DBMS Tables Normalization Lab

Question 1

Problem Statement:

Normalize the table up to 3NF.

Dataset:

OrderID	CustomerName	Product	Price
1	Anuj	Laptop	50000
1	Anuj	Mouse	500
2	Amit	Laptop	50000
2	Amit	Keyboard	1500
3	Ravi	Mouse	500
4	Sita	Laptop	50000
5	Mohan	Keyboard	1500
6	Riya	Mouse	500
7	Karan	Laptop	50000
8	Neha	Keyboard	1500

Answer :-

Order table: -

OrderID	CustomerName
1	Anuj
2	Amit
3	Ravi
4	Sita
5	Mohan
6	Riya
7	Karan
8	Neha

Product table:-

Product	Price
Laptop	50000
Mouse	500
Keyboard	1500

Order_Items table :-

OrderID	Product
1	Laptop
1	Mouse
2	Laptop
2	Keyboard
3	Mouse
4	Laptop
5	Keyboard
6	Mouse
7	Laptop
8	Keyboard

Question 2

Problem Statement:

Identify dependencies and normalize to 3NF.

Dataset:

EmpID	EmpName	DeptID	DeptName
101	**Anuj**	**D1**	**HR**
102	**Amit**	**D2**	**IT**
103	**Ravi**	**D1**	**HR**
104	**Sita**	**D3**	**Finance**
105	**Mohan**	**D2**	**IT**
106	**Riya**	**D1**	**HR**
107	**Karan**	**D3**	**Finance**

108	**Neha**	**D2**	**IT**
109	**Pooja**	**D1**	**HR**
110	**Rahul**	**D3**	**Finance**

Answer :-

Employee Table: -

EmpID	EmpName	DepID
101	Anuj	D1
102	Amit	D2
103	Ravi	D1
104	Sita	D3
105	Mohan	D2
106	Riya	D1
107	Karan	D3
108	Neha	D2
109	Pooja	D1
110	Rahul	D3

Department Table:-

DeptID	DeptName
D1	HR
D2	IT
D3	Finance

Question 3

Problem Statement:

Remove transitive dependency and normalize.

Dataset:

StudentID	Subject	Teacher	TeacherPhone
1	**Math**	**Ravi**	**9999**
1	**Science**	**Ravi**	**9999**

2	**English**	**Sita**	**8888**
3	**Math**	**Ravi**	**9999**
4	**Science**	**Ravi**	**9999**
5	**English**	**Sita**	**8888**
6	**Math**	**Ravi**	**9999**
7	**Science**	**Ravi**	**9999**
8	**English**	**Sita**	**8888**
9	**Math**	**Ravi**	**9999**

Answer :-

Student_Enrollment Table: -

StudentID	**Subject**	**Teacher**
1	Math	Ravi
1	Science	Ravi
2	English	Sita
3	Math	Ravi
4	Science	Ravi
5	English	Sita
6	Math	Ravi
7	Science	Ravi
8	English	Sita
9	Math	Ravi

Teacher Table: -

Teacher	**TeacherPhone**
Ravi	9999
Sita	8888

Question 4

Problem Statement:

Normalize to 3NF and identify all dependencies.

Dataset:

BookID	Author	Publisher	PublisherAddress
B1	A1	P1	Delhi
B2	A2	P2	Mumbai
B3	A1	P1	Delhi
B4	A3	P3	Pune
B5	A2	P2	Mumbai
B6	A1	P1	Delhi
B7	A3	P3	Pune
B8	A2	P2	Mumbai
B9	A1	P1	Delhi
B10	A3	P3	Pune

Answer :-

Books Table:-

BookID	Author	Publisher
B1	A1	P1
B2	A2	P2
B3	A1	P1
B4	A3	P3
B5	A2	P2
B6	A1	P1
B7	A3	P3
B8	A2	P2
B9	A1	P1
B10	A3	P3

Publisher Table:-

Publisher	PublisherAddress
P1	Delhi
P2	Mumbai

P3	pune

Question 5

Problem Statement:

Find candidate key and normalize to 3NF.

Dataset:

OrderID	ProductID	ProductName	CustomerID	CustomerName
1	P1	Laptop	C1	Anuj
1	P2	Mouse	C1	Anuj
2	P1	Laptop	C2	Amit
2	P3	Keyboard	C2	Amit
3	P2	Mouse	C3	Ravi
4	P1	Laptop	C4	Sita
5	P3	Keyboard	C5	Mohan
6	P2	Mouse	C6	Riya
7	P1	Laptop	C7	Karan
8	P3	Keyboard	C8	Neha

Answer :-

Customers Table :-

CustomerID	CustomerName
C1	Anuj
C2	Amit
C3	Ravi
C4	Sita
C5	Mohan
C6	Riya
C7	Karan
C8	Neha

Orders Table :-

OrderID	CustomerID
1	C1
2	C2
3	C3
4	C4
5	C5
6	C6
7	C7
8	C8

Products Table :-

ProductID	ProductName
P1	Laptop
P2	Mouse
P3	Keyboard

Order_Details Table :-

OrderID	ProductID
1	P1
1	P2
2	P1
2	P3
3	P2
4	P1
5	P3
6	P2
7	P1
8	P3

Question 6

Problem Statement:

Normalize the student-course table up to 3NF.

Dataset:

StudentID	StudentName	Course	Instructor
1	Anuj	Math	Ravi
1	Anuj	Science	Ravi
2	Amit	English	Sita
3	Ravi	Math	Ravi
4	Sita	Science	Ravi
5	Mohan	English	Sita
6	Riya	Math	Ravi
7	Karan	Science	Ravi
8	Neha	English	Sita
9	Pooja	Math	Ravi

Answer :-

Student Table :-

StudentID	StudentName
1	Sita
1	Mohan
2	Anuj
3	Anuj
4	Amit
5	Ravi
6	Riya
7	Karan
8	Neha
9	Pooja

Course Table :-

Course	Instructor
Math	Ravi
Science	Ravi
English	Sita

Enrollment Table :-

StudentID	Course
1	Math
1	Science
2	English
3	Math
4	Science
5	English
6	Math
7	Science
8	English
9	Math

Question 7

Problem Statement:

Normalize employee-project data to 3NF.

Dataset:

EmpID	ProjectID	ProjectName	Manager
101	P1	Alpha	Raj
101	P2	Beta	Raj
102	P1	Alpha	Raj
103	P3	Gamma	Simran
104	P2	Beta	Raj
105	P3	Gamma	Simran
106	P1	Alpha	Raj
107	P2	Beta	Raj
108	P3	Gamma	Simran
109	P1	Alpha	Raj

Answer :-

Projects Table :-

ProjectID	ProjectName	Manager
P1	Alpha	Raj
P2	Beta	Raj
P3	Gamma	Simran

Employee_Projects Table :-

EmpID	ProjectID
101	P1
101	P2
102	P1
103	P3
104	P2
105	P3
106	P1
107	P2
108	P3
109	P1

Question 8

Problem Statement:

Normalize sales data up to 3NF.

Dataset:

SaleID	Product	Category	Price
1	Laptop	Electronics	50000
2	Mouse	Electronics	500
3	Keyboard	Electronics	1500
4	Shirt	Clothing	1000
5	Jeans	Clothing	2000
6	Laptop	Electronics	50000
7	Mouse	Electronics	500

8	**Shirt**	**Clothing**	**1000**
9	**Jeans**	**Clothing**	**2000**
10	**Keyboard**	**Electronics**	**1500**

Answer :-

Sales Table :-

SaleID	Product
1	Laptop
2	Mouse
3	Keyboard
4	Shirt
5	Jeans
6	Laptop
7	Mouse
8	Shirt
9	Jeans
10	Keyboard

Products Table :-

Product	Category	Price
Laptop	Electronics	50000
Mouse	Electronics	500
Keyboard	Electronics	1500
Shirt	Clothing	1000
Jeans	Clothing	2000

Question 9

Problem Statement:

Normalize hospital patient records to 3NF.

Dataset:

PatientID	PatientName	Doctor	DoctorSpecialization

1	**Anuj**	**Dr. Ravi**	**Cardiology**
2	**Amit**	**Dr. Sita**	**Neurology**
3	**Ravi**	**Dr. Ravi**	**Cardiology**
4	**Sita**	**Dr. Sita**	**Neurology**
5	**Mohan**	**Dr. Ravi**	**Cardiology**
6	**Riya**	**Dr. Sita**	**Neurology**
7	**Karan**	**Dr. Ravi**	**Cardiology**
8	**Neha**	**Dr. Sita**	**Neurology**
9	**Pooja**	**Dr. Ravi**	**Cardiology**
10	**Rahul**	**Dr. Sita**	**Neurology**

Patient Table :-

PatientID	PatientName	Doctor
1	Anuj	Dr. Ravi
2	Amit	Dr. Sita
3	Ravi	Dr. Ravi
4	Sita	Dr. Sita
5	Mohan	Dr. Ravi
6	Riya	Dr. Sita
7	Karan	Dr. Ravi
8	Neha	Dr. Sita
9	Pooja	Dr. Ravi
10	Rahul	Dr. Sita

Doctor Table :-

Doctor	DoctorSpecialization
Dr. Ravi	Cardiology
Dr. Sita	Neurology

Question 10

Problem Statement:

Normalize university data to 3NF.

Dataset:

StudentID	Dept	HOD
1	CSE	Dr. A
2	IT	Dr. B
3	CSE	Dr. A
4	ECE	Dr. C
5	IT	Dr. B
6	CSE	Dr. A
7	ECE	Dr. C
8	IT	Dr. B
9	CSE	Dr. A
10	ECE	Dr. C

Answer :-

Student_Department Table :-

StudentID	Dept
1	CSE
2	IT
3	CSE
4	ECE
5	IT
6	CSE
7	ECE
8	IT
9	CSE
10	ECE

Department_HOD Table :-

Dept	HOD

CSE	Dr. A
IT	Dr. B
ECE	Dr. C